

Fig. 3: (a) Blue car moving towards stationary white car and going to collide, (b) the cars moving towards each other and going to collide, and (c) both the cars moving in the same direction

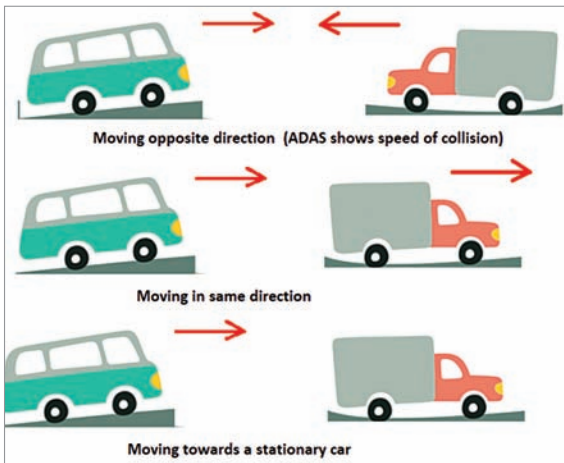


Fig. 4: The three different scenarios of cars movements

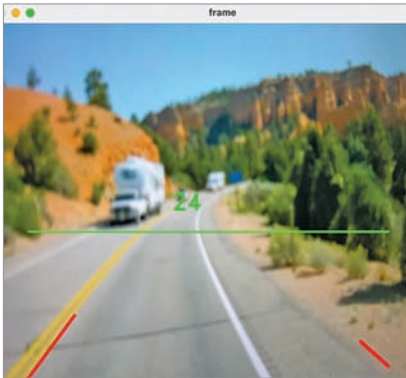


Fig. 5: ADAS display showing the safe distance marked line and speed at which both are approaching each other

pler radar that can work in all types of light conditions and foggy environment as well. It is also capable of measuring very high speeds that most of the other systems fail to do.

The components required to build this system are listed in the Bill of Materials table.

Fig. 2 shows the circuit diagram of the doppler radar in ADAS. It is

built around Raspberry Pi (MOD1), LM358 module, HB100 doppler radar, Raspberry Pi camera, and a few other components.

Let us first see how the doppler radar works, how it differs from the other radars, and how we implement it here.

Speed calculation using radar

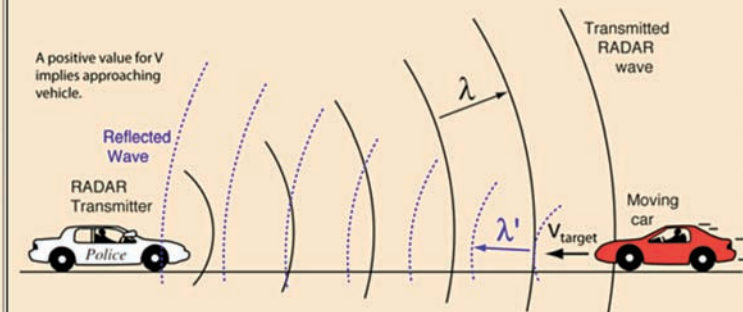
Doppler radar is based on the principle of doppler effect. Also known as the doppler shift, it describes the change in frequency or wavelength of a wave (such as sound or light) in relation to an observer moving the source of the wave.

BILL OF MATERIALS

Component	Quantity	Description
Raspberry Pi/Nvidia Jetson (MOD1)	1	SBC
Raspberry Pi camera	1	CSI camera
HB100 radar (S1)	1	10.525GHz radar sensor
LM358 amplifier (MOD2)	1	Signal amplifier module
5V DC-DC power adaptor	1	5V, 2A DC-DC power converter
Wires	10	Thin insulated wires

Beat Frequency and Speed

For a transmitted frequency of GHz = Hz
 the reflected frequency will be Hz
 and the beat frequency will be Hz
 if the speed of the vehicle is v_{target} = m/s = mi/hr = km/hr = ft/s



Note: If not entered as data, the transmitted frequency will default to 10.6 GHz, a typical X-band radar frequency. K-band units with frequencies on the order of 30 GHz are also used. Any parameters can be changed.

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Fig. 6: Speed calculator formula for doppler radar using the frequency (Source: HyperPhysics-radar)